Linear Algebra

Lectures 1–6

Definitions, Properties and Proofs

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Lecture map and notation

Lecture	Topics
1	Echelon forms, linear systems and span
2	Vector spaces, subspaces, bases and dimension
3	Linear maps, rank–nullity, one-to-one and onto maps
4	Coordinates, matrices, inverses and isomorphisms
5	Change of basis, duality and the double dual
6	Matrix rank, block operations, LU and inversion

Notation used throughout

$\mathbb{F}$ is a field. Vector spaces and linear maps use the same field. Ordered bases are written $\beta = (v_1, \ldots, v_n)$. In $[T]^{\gamma}_{\beta}$, the **subscript is the domain basis** and the **superscript is the codomain basis**.

Row echelon form and reduced row echelon form

定义 / Definition 1.1 Row echelon form

A **pivot** is the first nonzero entry of a nonzero row.

- Zero rows are below all nonzero rows.
- Pivots move strictly right as rows move down.
- Entries below each pivot are zero.

$$\begin{pmatrix} 2 & 1 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{pmatrix}$$

定义 / Definition 1.2 Reduced row echelon form

The matrix is in row echelon form, and:

- Every pivot equals 1.
- Every pivot is the only nonzero entry in its column.

$$\begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

性质 / Property 1.1 Existence and uniqueness

Every matrix is row equivalent to exactly one matrix in reduced row echelon form.

▸ Proof in appendix

Reading solutions from echelon form

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Reduce the **entire** augmented matrix $[\,A \mid b\,]$.

性质 / Property 1.2 Consistency and free variables

- A row $[\,0 \;\cdots\; 0 \mid c\,]$, $c \neq 0$, means no solution.
- Pivot columns **of** A give basic variables.
- The other columns give free variables.

If the system is consistent, it has a unique solution with no free variables, and infinitely many otherwise.

证明 / Proof Back substitution

Row operations preserve the solution set. Each choice of the free variables determines all basic variables uniquely.

□

例 / Example One free variable

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Basic variables: x_1, x_3.
Free variable: $x_2 = t \in \mathbb{R}$.

$$x = \begin{pmatrix} 3-2t \\ t \\ -1 \end{pmatrix}.$$

Span and existence of solutions

定义 / Definition 1.3 Span

For $v_1, \ldots, v_k \in V$,

$$\operatorname{span}\{v_1, \ldots, v_k\} = \left\{ \sum_{i=1}^{k} c_i v_i : c_i \in \mathbb{F} \right\}.$$

性质 / Property 1.3 Four equivalent statements

Let $A = [\, a_1 \ \cdots \ a_n \,] \in \mathbb{F}^{m \times n}$.

1. $Ax = b$ is solvable for every $b \in \mathbb{F}^m$.
2. Every $b \in \mathbb{F}^m$ is a linear combination of $a_1, \ldots, a_n$.
3. $\operatorname{span}\{a_1, \ldots, a_n\} = \mathbb{F}^m$.
4. A has a pivot in every row.

证明 / Proof Columns and consistency

Since $Ax = \sum_j x_j a_j$, statements 1–3 are equivalent.
If the row echelon form of A has a pivot in each row, back substitution works for every transformed right-hand side.
If a zero row occurs, choose the transformed right-hand side to be 1 in that row. Reverse the row operations to obtain a b making $Ax = b$ inconsistent.

□

Fields and vector spaces

定义 / Definition 2.1 Field

A field $\mathbb{F}$ has $0 \neq 1$, with:

- $(\mathbb{F}, +)$ an abelian group.
- $(\mathbb{F} \setminus \{0\}, \cdot)$ an abelian group.
- Multiplication distributive over addition.

Examples include $\mathbb{R}$, $\mathbb{C}$ and the rational numbers $\mathbb{Q}$.

定义 / Definition 2.2 Vector space over $\mathbb{F}$

A vector space is an abelian group $(V, +)$ with scalar multiplication $\mathbb{F} \times V \to V$ satisfying

$$a(u+v) = au + av, \quad (a+b)v = av + bv,$$
$$a(bv) = (ab)v, \qquad 1v = v.$$

These identities hold for all $a, b \in \mathbb{F}$ and $u, v \in V$.

Subspaces and the subspace test

定义 / Definition 2.3 Subspace

A subset $W \subseteq V$ is a subspace if it is a vector space under the addition and scalar multiplication inherited from V.

性质 / Property 2.1 Subspace test

$W \subseteq V$ is a subspace if and only if:

1. $0 \in W$.
2. $u + v \in W$ for all $u, v \in W$.
3. $cu \in W$ for all $c \in \mathbb{F}$, $u \in W$.

证明 / Proof Inherited operations

A subspace must satisfy all three conditions. Conversely, these conditions give zero, closure under addition, and additive inverses because $-u = (-1)u \in W$. Thus $(W, +)$ is an abelian group. All other vector-space identities are inherited from V. □

Intersections of subspaces

性质 / Property 2.2 Arbitrary intersections

If each W_i is a subspace of V, then $W = \bigcap_{i \in I} W_i$ is a subspace of V. For $I = \varnothing$, take the intersection to be V.

证明 / Proof Apply the subspace test

For a nonempty index set I, each W_i contains 0, so $0 \in W$. If $u, v \in W$ and $c \in \mathbb{F}$, then for every i,

$$u, v \in W_i \quad \Longrightarrow \quad u + v \in W_i, \qquad cu \in W_i.$$

Therefore $u + v, cu \in W$. The empty-intersection case is $W = V$.

□

例 / Example An arbitrary subset may fail the test

$\{(1, 0), (0, 1)\} \subseteq \mathbb{R}^2$ is not a subspace: it does not contain 0 and is not closed under addition.

Linear independence and bases

定义 / Definition 2.4 Linear independence

Vectors $v_1, \dots, v_k$ are linearly independent if

$$c_1 v_1 + \cdots + c_k v_k = 0 \quad \Longrightarrow \quad c_1 = \cdots = c_k = 0.$$

An infinite set is independent when every finite selection of distinct vectors from it is independent.

定义 / Definition 2.5 Basis

A basis is a linearly independent spanning set. A linear combination always has finitely many nonzero coefficients.

性质 / Property 2.3 Unique expansion

Every vector has a unique expression as a finite linear combination of basis vectors.

证明 / Proof Spanning and independence

Spanning gives existence. If $\sum_i a_i v_i = \sum_i b_i v_i$, subtract to obtain

$$\sum_i (a_i - b_i) v_i = 0.$$

Independence gives $a_i = b_i$ for every i.

□

The exchange lemma

性质 / Property 2.4 An independent list fits inside a spanning list

Suppose $u_1, \dots, u_m$ are independent and $s_1, \dots, s_n$ span V. Then $m \le n$. We can replace m of the s_j by $u_1, \dots, u_m$ and still obtain a spanning list of n vectors.

证明 / Proof Successive replacement

Suppose a spanning list already contains $u_1, \dots, u_{k-1}$. Expand u_k in that list:

$$u_k = \sum_{i<k} a_i u_i + \sum_{j \text{ remaining}} b_j s_j.$$

Some b_j is nonzero, since otherwise $u_1, \dots, u_k$ would be dependent. Solve for that s_j and replace it by u_k. The new list still spans V.

Each replacement consumes one of the n original slots. After n replacements the first n vectors u_i span V, so an additional u_{n+1} would violate independence. Hence $m \le n$.

□

Dimension and the basis criteria

性质 / Property 2.5　All finite bases have the same size

If V has a basis of n vectors, every independent set has at most n vectors, and every spanning set has at least n vectors. Consequently, every basis has exactly n vectors.

定义 / Definition 2.6　Dimension

This common size is $\dim V$. The zero space has the empty basis and dimension 0.

性质 / Property 2.6　A list of $n = \dim V$ vectors

- An independent list of length n is a basis.
- A spanning list of length n is a basis.

证明 / Proof　Consequences of the exchange lemma

Apply Property 2.4 to a basis and the other list. If an independent list of length n failed to span, one could append a vector and keep independence. If a spanning list of length n were dependent, one could remove a vector and keep spanning. Both conclusions violate the size bounds.

□

Linear maps, range and null space

定义 / Definition 3.1 Linear map

A map $T\colon V \to W$ is linear if $T(au+bv) = aT(u) + bT(v)$ for all $a, b \in \mathbb{F}$ and $u, v \in V$.

定义 / Definition 3.2 Range and null space

$$\mathrm{R}(T) = \{T(v) : v \in V\} \subseteq W, \qquad \mathrm{N}(T) = \{v \in V : T(v) = 0\} \subseteq V.$$

性质 / Property 3.1 Both are subspaces

$\mathrm{R}(T)$ is a subspace of W, and $\mathrm{N}(T)$ is a subspace of V.

证明 / Proof Linearity gives closure

$T(0) = 0$. If $T(u), T(v) \in \mathrm{R}(T)$, then $aT(u) + bT(v) = T(au+bv) \in \mathrm{R}(T)$. If $u, v \in \mathrm{N}(T)$, then $T(au+bv) = 0$. Apply the subspace test in each case. □

The rank–nullity theorem

定义 / Definition 3.3 Rank and nullity

For a linear map $T\colon V \to W$ with V finite-dimensional,

$$\operatorname{rank} T = \dim \mathrm{R}(T), \qquad \operatorname{nullity} T = \dim \mathrm{N}(T).$$

性质 / Property 3.2 Rank–nullity

If V is finite-dimensional, then

$$\boxed{\dim V = \operatorname{nullity} T + \operatorname{rank} T.}$$

The codomain W need not be finite-dimensional.

例 / Example A map from $\mathbb{R}^3$ to $\mathbb{R}^2$

Let $T(x, y, z) = (x + y, z)$. Then

$$\mathrm{N}(T) = \operatorname{span}\{(1, -1, 0)\}, \qquad \mathrm{R}(T) = \mathbb{R}^2.$$

Thus $\operatorname{nullity} T = 1$, $\operatorname{rank} T = 2$, and $1 + 2 = 3 = \dim \mathbb{R}^3$.

Proof of rank–nullity

证明 / Proof Property 3.2

Choose a basis $u_1, \dots, u_k$ of $\mathrm{N}(T)$ and extend it to a basis

$$u_1, \dots, u_k, v_1, \dots, v_r \quad \text{of } V.$$

Such an extension exists by repeatedly adding a vector outside the current span until the list spans V.
Spanning. Applying T to a basis expansion of any $x \in V$ gives $T(x) \in \operatorname{span}\{T(v_1), \dots, T(v_r)\}$.
Independence. If $\sum_j c_j T(v_j) = 0$, then $\sum_j c_j v_j \in \mathrm{N}(T)$, so $\sum_j c_j v_j = \sum_i a_i u_i$. Independence of the extended basis gives every $c_j = 0$.
Therefore $T(v_1), \dots, T(v_r)$ form a basis of $\mathrm{R}(T)$, and

$$\dim V = k + r = \text{nullity } T + \text{rank } T.$$

□

One-to-one and onto linear maps

性质 / Property 3.3 Kernel criterion

For any linear $T\colon V \to W$,

$$T \text{ is one-to-one} \iff \mathrm{N}(T) = \{0\}.$$

证明 / Proof Compare two inputs

If T is one-to-one, $T(x) = T(0)$ gives $x = 0$. Conversely, if $T(u) = T(v)$, then $T(u - v) = 0$. A trivial null space forces $u = v$.

□

性质 / Property 3.4 Equal finite dimensions

If $\dim V = \dim W = n < \infty$, then

$$\begin{aligned} T \text{ one-to-one} &\iff \operatorname{rank} T = n \\ &\iff T \text{ onto.} \end{aligned}$$

证明 / Proof Use rank–nullity

Being one-to-one means $\operatorname{nullity} T = 0$, equivalently $\operatorname{rank} T = n$. A subspace of the n-dimensional space W has dimension n exactly when it equals W.

□

One-to-one maps preserve and reflect independence

性质 / Property 3.5 Independence under a one-to-one linear map

Let $T\colon V \to W$ be one-to-one and linear. A set $S \subseteq V$ is linearly independent if and only if $T(S)$ is linearly independent.

证明 / Proof Apply T to a finite relation

Preservation. Suppose $v_1, \dots, v_k \in S$ are distinct and $\sum_i c_i T(v_i) = 0$. Then

$$T\left(\sum_i c_i v_i\right) = 0 \quad \Longrightarrow \quad \sum_i c_i v_i = 0 \quad \Longrightarrow \quad c_i = 0 \text{ for all } i.$$

Reflection. If $\sum_i c_i v_i = 0$, applying T gives $\sum_i c_i T(v_i) = 0$. Since T is one-to-one, the $T(v_i)$ are distinct, so independence of $T(S)$ gives every $c_i = 0$.
The argument covers infinite S, since every relation is finite.

□

A linear map is determined by its values on a basis

性质 / Property 3.6 Existence and uniqueness

Given a basis $(v_1, \ldots, v_n)$ of V and arbitrary $w_1, \ldots, w_n \in W$, there is a unique linear map $T\colon V \to W$ such that $T(v_i) = w_i$. It is given by

$$T\left(\sum_{i=1}^{n} c_i v_i\right) = \sum_{i=1}^{n} c_i w_i.$$

证明 / Proof Define the map using unique coordinates

Every input has unique coefficients c_i, so the formula is well-defined. Adding inputs or multiplying an input by a scalar adds or scales those coefficients, proving linearity. The formula gives $T(v_i) = w_i$. Any linear map with those basis values must satisfy the same formula, proving uniqueness.

□

No equality between $\dim V$ and $\dim W$ is required.

Coordinate representation of a vector

定义 / Definition 4.1 Coordinates in an ordered basis

Let $\beta = (v_1, \ldots, v_n)$ be an ordered basis of V. If $v = c_1 v_1 + \cdots + c_n v_n$, define

$$[v]_\beta = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \in \mathbb{F}^n.$$

The order of the basis vectors determines the order of the coordinates.

性质 / Property 4.1 The coordinate map is one-to-one and onto

$\phi_\beta : V \to \mathbb{F}^n$, $\phi_\beta(v) = [v]_\beta$, is linear, one-to-one and onto.

证明 / Proof Unique basis expansions

Coordinates add and scale with vectors, so ϕ_β is linear. Unique expansions make it one-to-one. Every $c \in \mathbb{F}^n$ is the image of $\sum_i c_i v_i$, so it is onto.

□

The matrix of a linear map

定义 / Definition 4.2 Left multiplication and matrix representation

For $A \in \mathbb{F}^{m\times n}$, the map $L_A : \mathbb{F}^n \to \mathbb{F}^m$, $L_A(x) = Ax$, is linear.
For $T\colon V \to W$, with bases $\beta = (v_1, \dots, v_n)$ and $\gamma = (w_1, \dots, w_m)$, define

$$A = [T]^{\gamma}_{\beta} = \big[\, [T(v_1)]_\gamma \ \cdots \ [T(v_n)]_\gamma \,\big].$$

性质 / Property 4.2 The coordinate identity

For every $v \in V$, $\boxed{[T(v)]_\gamma = [T]^{\gamma}_{\beta}[v]_\beta}$.

证明 / Proof Expand the input

Write $v = \sum_j c_j v_j$. Then $T(v) = \sum_j c_j T(v_j)$, so $[T(v)]_\gamma = \sum_j c_j [T(v_j)]_\gamma = A[v]_\beta$.

□

Invertibility and linearity of the inverse

定义 / Definition 4.3 Inverse map

$T\colon V \to W$ is invertible if a map $T^{-1}\colon W \to V$ satisfies

$$T^{-1} \circ T = \mathrm{id}_V, \qquad T \circ T^{-1} = \mathrm{id}_W.$$

性质 / Property 4.3 Invertible linear maps

A linear map is invertible if and only if it is both one-to-one and onto. Its inverse is linear.

证明 / Proof Unique preimages

The two inverse identities show that T is one-to-one and onto. Conversely, if T is one-to-one and onto, every $y \in W$ has a unique preimage. This defines $T^{-1}(y)$ and satisfies both identities.
For $y, z \in W$ and $a, b \in \mathbb{F}$,

$$T\big(aT^{-1}(y) + bT^{-1}(z)\big) = ay + bz.$$

Uniqueness of the preimage gives $T^{-1}(ay + bz) = aT^{-1}(y) + bT^{-1}(z)$.

□

The matrix of an inverse

Let $T\colon V \to W$ be invertible, with V, W finite-dimensional. Let β be a basis of V and γ a basis of W.

性质 / Property 4.4 The bases exchange roles

If $A = [T]_\beta^\gamma$, then

$$\boxed{[T^{-1}]_\gamma^\beta = A^{-1}.}$$

For T^{-1}, the domain basis is γ and the codomain basis is β.

证明 / Proof Use the two inverse identities

Since T is one-to-one and onto, rank–nullity gives $\dim V = \dim W$, so A is square. Write $B = [T^{-1}]_\gamma^\beta$. For every $v \in V$ and $w \in W$,

$$BA[v]_\beta = [T^{-1}(T(v))]_\beta = [v]_\beta, \qquad AB[w]_\gamma = [w]_\gamma.$$

Hence $BA = I$ and $AB = I$, which gives $B = A^{-1}$.

□

Isomorphisms preserve vector-space structure

定义 / Definition 4.4 Isomorphism

An isomorphism is an invertible linear map. We write $V \cong W$ when an isomorphism $V \to W$ exists.

性质 / Property 4.5 What an isomorphism preserves

If $T\colon V \to W$ is an isomorphism, then:

- T maps bases to bases and preserves and reflects independence.
- For a subspace $U \subseteq V$, $T(U)$ is a subspace of W and $T|_U : U \to T(U)$ is an isomorphism.

In finite dimensions, $V \cong W$ if and only if $\dim V = \dim W$.

证明 / Proof One-to-one and onto maps

A one-to-one map preserves independence, and an onto map sends a spanning set onto a spanning set. Linearity gives closure of $T(U)$, and the restriction is one-to-one and has range $T(U)$. Conversely, bases of the same finite size define an isomorphism by mapping corresponding basis vectors to one another.

□

The vector space of linear maps

定义 / Definition 4.5 The space $\mathcal{L}(V, W)$

$\mathcal{L}(V, W)$ is the vector space of linear maps $V \to W$, with pointwise operations:

$$(S+T)(v) = S(v) + T(v), \qquad (aT)(v) = aT(v).$$

性质 / Property 4.6 Dimension of the map space

If $\dim V = n$ and $\dim W = m$ are finite, then

$$\boxed{\dim \mathcal{L}(V, W) = mn.}$$

证明 / Proof Identify maps with matrices

Fix bases β, γ. The assignment $T \mapsto [T]_\beta^\gamma$ is linear. It is one-to-one and onto because arbitrary matrix columns prescribe arbitrary images of the basis vectors (Property 3.6).
The space $\mathbb{F}^{m\times n}$ has a basis of mn matrices, each with one entry equal to 1 and all other entries zero. Therefore its dimension, and that of $\mathcal{L}(V, W)$, is mn.

□

Change of coordinates: which direction does Q act?

定义 / Definition 5.1 Change-of-coordinate matrix

For ordered bases β and $\beta' = (v'_1, \dots, v'_n)$ of V, define

$$Q = [\mathrm{id}_V]_{\beta'}^{\beta} = \big[\, [v'_1]_\beta \ \cdots \ [v'_n]_\beta \,\big].$$

性质 / Property 5.1 Coordinate conversion

Q is invertible, and $\boxed{[v]_\beta = Q[v]_{\beta'}}$. Thus Q converts β'-**coordinates into** β-**coordinates**.

证明 / Proof Apply the coordinate identity to id_V

Property 4.2 gives $[\mathrm{id}_V(v)]_\beta = Q[v]_{\beta'}$. Since id_V is invertible, the inverse conversion matrix is $Q^{-1} = [\mathrm{id}_V]_{\beta}^{\beta'}$.

□

例 / Example In $\mathbb{R}^2$

If $\beta = (e_1, e_2)$ and $\beta' = (e_1, e_1 + e_2)$, then $Q = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Its columns are the β-coordinates of the vectors in β'.

Similarity represents the same operator in two bases

性质 / Property 5.2 Change of basis for a linear operator

Let $T\colon V \to V$, $A = [T]_{\beta}^{\beta}$, $A' = [T]_{\beta'}^{\beta'}$, and $Q = [\mathrm{id}_V]_{\beta'}^{\beta}$. Then

$$\boxed{A' = Q^{-1} A Q.}$$

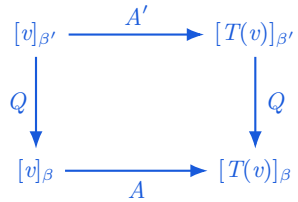
Two square matrices related in this way are called **similar**.

证明 / Proof Convert before and after applying T

For every $v \in V$,

$$QA'[v]_{\beta'} = [T(v)]_{\beta},$$
$$AQ[v]_{\beta'} = A[v]_{\beta} = [T(v)]_{\beta}.$$

Hence $QA' = AQ$. Multiplying by Q^{-1} on the left gives the result. □



Both routes give the same coordinates.

Dual spaces and dual bases

定义 / Definition 5.2 Dual space and dual basis

The dual space is $V^* = \mathcal{L}(V, \mathbb{F})$, the space of linear functionals. Given a basis $\beta = (v_1, \dots, v_n)$, define its **dual basis** $\beta^* = (f_1, \dots, f_n)$ by

$$f_i\left(\sum_j c_j v_j\right) = c_i, \qquad f_i(v_j) = \delta_{ij}.$$

性质 / Property 5.3 Expansion in the dual basis

If $\dim V = n < \infty$, then for every $f \in V^*$,

$$f = \sum_{i=1}^{n} f(v_i) f_i, \qquad \dim V^* = \dim V = n.$$

证明 / Proof Evaluate on basis vectors

Both sides take the same value on every v_j, so they are equal by Property 3.6. If $\sum_i a_i f_i = 0$, evaluating on v_j gives $a_j = 0$. Thus the f_i span V^* and are independent.

□

Dual maps and transpose matrices

定义 / Definition 5.3 Dual map

For a linear map $T: V \to W$, define

$$T^t: W^* \to V^*, \qquad T^t(g) = g \circ T.$$

It is linear because $(ag + bh) \circ T = a(g \circ T) + b(h \circ T)$.

性质 / Property 5.4 The matrix in dual bases

Let V, W be finite-dimensional, with bases β, γ and dual bases β^*, γ^*. If $A = [T]_\beta^\gamma$, then

$$\boxed{[T^t]_{\gamma^*}^{\beta^*} = A^{\mathsf{T}}.}$$

If $\dim V = n$ and $\dim W = m$, then A is $m \times n$, while $[T^t]_{\gamma^*}^{\beta^*}$ is $n \times m$.

Here T denotes the ordinary transpose, including over $\mathbb{C}$.

Proof of the transpose formula

证明 / Proof Property 5.4

Write $\beta = (v_i)$, $\gamma = (w_j)$, $\beta^* = (f_i)$ and $\gamma^* = (g_j)$. Let $B = [T^t]_{\gamma^*}^{\beta^*}$.
Column j of B expands $T^t(g_j)$ in the f_i:

$$T^t(g_j) = \sum_{i=1}^{n} B_{ij} f_i.$$

Evaluating at v_i gives

$$\begin{aligned} B_{ij} &= (T^t(g_j))(v_i) \\ &= g_j(T(v_i)) \\ &= A_{ji}. \end{aligned}$$

Thus $B = A^{\mathsf{T}}$.

□

例 / Example Dimensions exchange

For $T : \mathbb{F}^2 \to \mathbb{F}^3$,

$$A = \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}.$$

The dual map goes from $(\mathbb{F}^3)^*$ to $(\mathbb{F}^2)^*$, with matrix

$$A^{\mathsf{T}} = \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix}.$$

The double dual space and its canonical isomorphism

定义 / Definition 5.4 Double dual space and evaluation functionals

The double dual space is $V^{**} = (V^*)^* = \mathcal{L}(V^*, \mathbb{F})$. For each $x \in V$, define $\hat{x} : V^* \to \mathbb{F}$ by

$$\boxed{\hat{x}(f) = f(x) \qquad (f \in V^*).}$$

Evaluation is linear in f, so $\hat{x} \in V^{**}$.

性质 / Property 5.5 The canonical isomorphism

If V is finite-dimensional, the map $V \to V^{**}$, $x \mapsto \hat{x}$, is an isomorphism. Thus $V \cong V^{**}$.

证明 / Proof Linearity, the one-to-one property and dimension

For every $f \in V^*$, $\widehat{ax + by}(f) = af(x) + bf(y) = (a\hat{x} + b\hat{y})(f)$, so $x \mapsto \hat{x}$ is linear. If $\hat{x} = 0$, evaluating on each dual-basis functional f_i gives $f_i(x) = 0$. All coordinates of x vanish, so $x = 0$.
The map is one-to-one. Since $\dim V^{**} = \dim V < \infty$, it is also onto.

□

The formula $\hat{x}(f) = f(x)$ uses no chosen basis.

Matrix rank and elementary row operations

定义 / Definition 6.1 Matrix rank

For $A \in \mathbb{F}^{m\times n}$,

$$\operatorname{Col}(A) = \operatorname{span}\{\text{columns of } A\}, \qquad \operatorname{rank} A = \dim \operatorname{Col}(A) = \operatorname{rank} L_A.$$

性质 / Property 6.1 Row operations preserve rank

Each elementary row operation is left multiplication by an invertible elementary matrix E. Therefore

$$\boxed{\operatorname{rank}(EA) = \operatorname{rank} A.}$$

证明 / Proof An invertible map acts on the column space

Swapping rows, multiplying a row by a nonzero scalar, and adding a multiple of one row to another all have inverse row operations, so their matrices are invertible.
Since $\operatorname{Col}(EA) = E(\operatorname{Col}(A))$ and $x \mapsto Ex$ is an isomorphism, the two column spaces have the same dimension.

□

Invertible right multiplication preserves rank

性质 / Property 6.2 Right multiplication

Let $A \in \mathbb{F}^{m\times n}$ and let $B \in \mathbb{F}^{n\times n}$ be invertible. Then

$$\mathrm{Col}(AB) = \mathrm{Col}(A), \qquad \boxed{\mathrm{rank}(AB) = \mathrm{rank}\, A.}$$

证明 / Proof Show both inclusions

Every vector in $\mathrm{Col}(AB)$ has the form $ABx = A(Bx)$ and lies in $\mathrm{Col}(A)$.
Conversely, every vector $Ay \in \mathrm{Col}(A)$ can be written as $Ay = AB(B^{-1}y)$, so it lies in $\mathrm{Col}(AB)$. Thus the column spaces are equal and have equal dimension.

□

Operation	Column-space relation	Rank
$A \mapsto EA$, E invertible	$\mathrm{Col}(EA) = E(\mathrm{Col}(A))$	Unchanged
$A \mapsto AB$, B invertible	$\mathrm{Col}(AB) = \mathrm{Col}(A)$	Unchanged

Partitioned matrices and block multiplication

定义 / Definition 6.2 Compatible block partitions

Partition matrices into submatrices called **blocks**. Addition is componentwise for matching partitions. For multiplication, let A_{ij} have size $m_i \times k_j$ and $B_{j\ell}$ have size $k_j \times n_\ell$.

性质 / Property 6.3 The usual multiplication rule applies to blocks

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

证明 / Proof Split the entrywise sum at the block boundary

For an entry of $C = AB$, write

$$C_{rs} = \sum_{t=1}^{k_1+k_2} A_{rt}B_{ts} = \sum_{t=1}^{k_1} A_{rt}B_{ts} + \sum_{t=k_1+1}^{k_1+k_2} A_{rt}B_{ts}.$$

These two sums are exactly the two block products in the corresponding block.

□

LU decomposition from Gaussian elimination

定义 / Definition 6.3 LU decomposition

A factorization $A = LU$ has L unit lower triangular and U upper triangular. Rectangular versions use an upper trapezoidal U.

性质 / Property 6.4 Elimination without row exchanges

For a nonsingular square A, if Gaussian elimination needs no row exchanges, its multipliers give $A = LU$. An operation $R_i \leftarrow R_i - m_{ij}R_j$ records $L_{ij} = m_{ij}$.

证明 / Proof Reverse the elimination operations

If $E_r \cdots E_1 A = U$, then

$$A = E_1^{-1} \cdots E_r^{-1} U = LU.$$

The inverse operations add back the recorded multiples of earlier rows. Their product is unit lower triangular; in elimination order, its entries below the diagonal are the recorded multipliers.

□

With row pivoting, use $PA = LU$ and apply each row exchange to the previously recorded multipliers.

LU decomposition: a worked example

例 / Example Record the elimination multipliers

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \xrightarrow{m_{21}=2,\ m_{31}=4} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix} \xrightarrow{m_{32}=3} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = U.$$

The multipliers form L

$m_{21} = 2,\ m_{31} = 4,\ m_{32} = 3.$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}.$$

证明 / Proof Direct verification

Denote the rows of U by u_1, u_2, u_3. The rows of LU are

$$\begin{aligned} u_1 &= (2, 1, 1), \\ 2u_1 + u_2 &= (4, 3, 3), \\ 4u_1 + 3u_2 + u_3 &= (8, 7, 9). \end{aligned}$$

These are the rows of A, so $LU = A$.

□

Computing an inverse by row reduction

性质 / Property 6.5 Augment with the identity

For a square matrix A, apply row operations to $[\,A \mid I\,]$. If the left block becomes I, the right block is A^{-1}:

$$[\,A \mid I\,] \longrightarrow [\,I \mid A^{-1}\,].$$

If the left block cannot be reduced to I, then A is not invertible.

证明 / Proof Track the same row operations on both blocks

A product E of elementary matrices transforms the augmented matrix into $[\,EA \mid E\,]$. If $EA = I$, the square matrix A is invertible and $E = A^{-1}$. Conversely, an invertible A has full rank, so its reduced row echelon form is I.

□

例 / Example A 2×2 inverse

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right).$$

Key statements and their hypotheses

Statement	Hypotheses and conventions
$\dim V = \text{nullity } T + \text{rank } T$	$T\colon V \to W$ linear; $\dim V < \infty$.
One-to-one $\iff$ onto	$\dim V = \dim W < \infty$.
$[T^{-1}]_\gamma^\beta = ([T]_\beta^\gamma)^{-1}$	T invertible; ordered domain and codomain bases exchange roles.
$A' = Q^{-1}AQ$	$T : V \to V$; Q converts β'-coordinates to β-coordinates.
$[T^t]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^{\mathsf{T}}$	Finite dimensions; the specified dual bases; ordinary transpose.
$x \mapsto \hat{x}$ gives $V \cong V^{**}$	$\dim V < \infty$; $\hat{x}(f) = f(x)$.
$A = LU$, or $PA = LU$ with pivoting	Elimination without row exchanges, or with consistent row pivoting.

Exercise 1: a sum of inverse matrices

习题 / Exercise 1

Suppose A, B and $A+B$ are all invertible $n \times n$ matrices. Prove that $A^{-1}+B^{-1}$ is also invertible and find its inverse.

证明 / Proof Factor the sum

Observe that

$$A^{-1}+B^{-1} = A^{-1}(A+B)B^{-1} = B^{-1}(A+B)A^{-1}.$$

Each factor is invertible, so $A^{-1}+B^{-1}$ is invertible. Taking the inverse of the product reverses the order of the factors:

$$\boxed{\left(A^{-1}+B^{-1}\right)^{-1} = B(A+B)^{-1}A = A(A+B)^{-1}B.}$$

Either expression is the required inverse.

□

Exercise 2: a matrix satisfying $A^2 = 0$

习题 / Exercise 2

Let $A \in \mathbb{F}^{n\times n}$ have rank r, where $1 \le r \le n/2$, and suppose $A^2 = 0$. Prove that there exists an invertible matrix P such that

$$A = PNP^{-1}, \qquad N = \begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The row-block sizes of N are $r, n-r$ and the column-block sizes are $r, r, n-2r$. Omit the last column block when $n = 2r$.

证明 / Proof Step 1: the range lies in the null space

Set $T = L_A : \mathbb{F}^n \to \mathbb{F}^n$. Since $T^2 = 0$,

$$y = T(x) \quad \Longrightarrow \quad T(y) = T^2(x) = 0, \qquad \text{so } \mathrm{R}(T) \subseteq \mathrm{N}(T).$$

Rank–nullity gives $\dim \mathrm{R}(T) = r$ and $\dim \mathrm{N}(T) = n - r$. In particular, $r \le n - r$, which explains the condition $r \le n/2$.

□

Exercise 2: construct the required basis

证明 / Proof Step 2: choose null-space vectors and preimages

1. Choose a basis $u_1, \dots, u_r$ of $\mathrm{R}(T)$. Since $\mathrm{R}(T) \subseteq \mathrm{N}(T)$, every u_i satisfies $T(u_i) = 0$.
2. Extend this list to a basis of $\mathrm{N}(T)$:

   $$u_1, \dots, u_r, z_1, \dots, z_{n-2r}.$$

   Each z_j also satisfies $T(z_j) = 0$.
3. For each i, choose $v_i \in \mathbb{F}^n$ such that

   $$T(v_i) = u_i.$$

   Such a preimage exists because $u_i \in \mathrm{R}(T)$.

Consider the ordered list of n vectors

$$\beta = (u_1, \dots, u_r,\ v_1, \dots, v_r,\ z_1, \dots, z_{n-2r}).$$

We will show that β is a basis and use it to form P.

□

Exercise 2: verify the basis and the matrix form

证明 / Proof Step 3: prove that β is a basis

Suppose $\sum_i a_i u_i + \sum_i b_i v_i + \sum_j c_j z_j = 0$. Applying T gives $\sum_i b_i u_i = 0$, so every $b_i = 0$. The remaining relation is among the basis vectors u_i, z_j of $\mathrm{N}(T)$, so every $a_i = c_j = 0$. Thus the n vectors in β are independent and form a basis of $\mathbb{F}^n$.

□

证明 / Proof Step 4: write the matrix in this basis

Put the basis vectors into the columns of the invertible matrix

$$P = [\, u_1 \ \cdots \ u_r \ v_1 \ \cdots \ v_r \ z_1 \ \cdots \ z_{n-2r} \,].$$

Since $Au_i = 0$, $Av_i = u_i$ and $Az_j = 0$,

$$AP = [\, 0 \ \cdots \ 0 \ u_1 \ \cdots \ u_r \ 0 \ \cdots \ 0 \,] = P \begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \end{pmatrix} = PN.$$

Hence $\boxed{P^{-1}AP = N}$, or equivalently $\boxed{A = PNP^{-1}}$.

□

Recommended reference



推荐参考书 / Recommended reference

高等代数
典型问题与方法

樊启斌　编著

高等教育出版社

Supplementary reading and further exercises alongside Lectures 1–6.

谢谢大家

Thanks!

祝大家在 11.13 考出好成绩！

Appendix: existence of reduced row echelon form

Here $\mathrm{Row}(R)$ denotes the span of the rows of R.

证明 / Proof Property 1.1: existence and a row-space invariant

Existence. Gaussian elimination produces row echelon form. Scale each pivot to 1 and eliminate the entries above it, working from the last pivot upward. The result is in reduced row echelon form.

Pivot columns are determined by the row space. Let $r_1, \dots, r_k$ be the nonzero rows of a matrix R in reduced row echelon form, with pivot columns $p_1 < \cdots < p_k$. Every nonzero $x \in \mathrm{Row}(R)$ has the form

$$x = \sum_{i=1}^{k} c_i r_i.$$

If i is the smallest index with $c_i \neq 0$, then the first nonzero entry of x is in column p_i, where its value is c_i. Conversely, each p_i occurs as the first nonzero position of r_i.

Hence the pivot columns are exactly the possible first nonzero positions of vectors in the row space. They depend only on that space.

□

Appendix: uniqueness of reduced row echelon form

证明 / Proof Property 1.1: uniqueness

Suppose R and S are row-equivalent matrices in reduced row echelon form. Row operations preserve the row space, so

$$\mathrm{Row}(R) = \mathrm{Row}(S).$$

By the preceding argument, R and S have the same pivot columns $p_1 < \cdots < p_k$ and the same number k of nonzero rows.

For each i, compare the ith nonzero rows r_i and s_i. Both have entry δ_{ij} in column p_j. Thus $r_i - s_i$ has zero entries in every pivot column.

But $r_i - s_i$ lies in the common row space. Write $r_i - s_i = \sum_j c_j r_j$. Its entry in column p_j is c_j, so every $c_j = 0$. Therefore $r_i = s_i$ for every i. The remaining rows are zero, proving $R = S$.

□

◂ Row echelon forms